

1 · WORDS YOU ARE MARKED ON

Aetiology	the CAUSE
Pathogenesis	the MECHANISM: cause → lesion
Sequelae	what follows
Idiopathic	cause unknown
Iatrogenic	caused by the vet
Nosocomial	hospital-acquired
Pathognomonic	lesion diagnostic of ONE disease
Morbidity	% affected Mortality % dying of total
Case fatality	% dying of those affected

NAMES ALWAYS ASKED Virchow — father of **modern/cellular** pathology; added **functio laesa**. **Celsus** — first 4 cardinal signs. **Morgagni** — organ pathology. **Bichat** — tissue. **Hippocrates** — humoral.

2 · CAUSES OF DISEASE

INTRINSIC — genetic (lethal/sub-lethal genes, freemartin 60,XX/XY chimera); immunological; age · sex · species · breed.
EXTRINSIC — physical (trauma, heat, cold, radiation, electricity, pressure); chemical (poisons, drugs, metals); nutritional (deficiency, excess, imbalance); biological (virus, bacteria, fungi, protozoa, helminth, arthropod, prion).

ONE-LINER A **predisposing** factor makes disease likely; the **determining/specific** cause produces it.

3 · BRANCHES & COURSE OF DISEASE

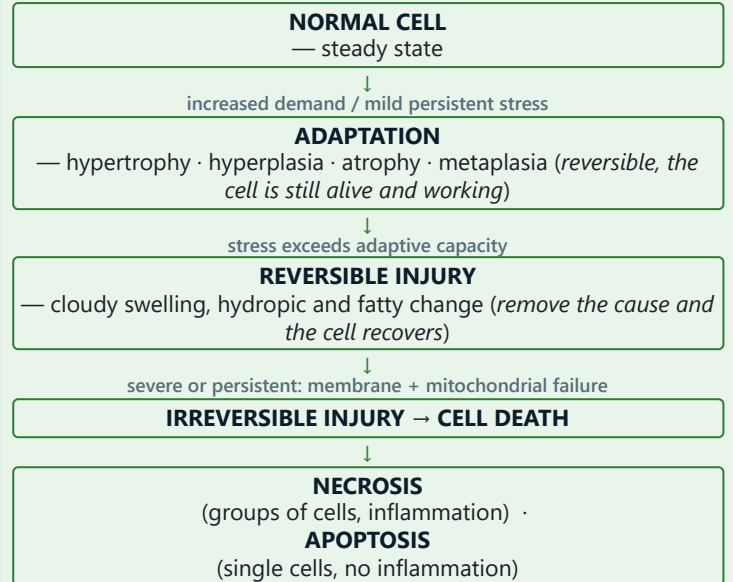
Branches — **General** (basic processes) · **Systemic/Special** (organ-wise) · **Clinical** (blood, urine, cytology) · **Surgical** (biopsy) · **Experimental** · **Comparative** · **Forensic/veterolegal**.

Course	Meaning	Course	Meaning
Peracute	Hours, often found dead	Subacute	Days–weeks
Acute	Sudden, short, severe	Chronic	Weeks–months, insidious

Incubation = infection → first sign · **Prodromal** = vague early signs · **Syndrome** = group of signs occurring together · **Lesion** = structural change · **Symptom** (felt) vs **Sign** (observed — all veterinary findings are signs).

DIAGNOSIS LADDER **Morphological** dx (e.g. "acute diffuse fibrinous pneumonia") → **Aetiological** dx (e.g. "pasteurellosis") → **Disease** dx ("haemorrhagic septicaemia").

4 · THE MASTER CHART — CELL UNDER STRESS



SIX CAUSES OF CELL INJURY Hypoxia / ischaemia (the commonest) · physical agents · chemicals and drugs · infectious agents · immunological reactions · genetic and nutritional derangements.

Hypoxia = low oxygen but blood is still flowing. **Ischaemia** = loss of blood supply, so oxygen **and** substrate are cut off and waste accumulates — therefore **ischaemia injures faster than hypoxia**.

5 · HYPERAEMIA VS CONGESTION

	Hyperaemia	Congestion
Nature	ACTIVE	PASSIVE
Mechanism	Arteriolar dilation , ↑inflow	Impaired venous outflow
Blood / colour	Oxygenated, bright red	Deoxygenated, dark blue-red
Temperature	Warm	Cool
Example	Inflammation, exercise	Heart failure, torsion, hypostatic

CHRONIC PASSIVE CONGESTION Liver = "nutmeg liver" (centrilobular congestion + necrosis). **Lung** = "brown induration" with **heart-failure cells** = haemosiderin-laden alveolar macrophages, **Perls' positive**. Spleen = cyanotic induration.

6 · HAEMORRHAGE

Term	Size	Term	Meaning
Petechiae	<3 mm	Suffusion	Diffuse sheet
Purpura	3 mm–1 cm	Haematoma	Confined mass
Ecchymosis	>1 cm	Apoplexy	Massive, into organ

Mechanism — **only two**: **per rhexis** = actual **rupture**; **per diapedesis** = RBCs through an **intact** wall.

Epistaxis nose · **Haemoptysis** coughed · **Haematemesis** vomited
Melena black digested · **Haematochezia** fresh · **Haematuria**
Haemo-thorax/pericardium/peritoneum · **Metrorrhagia** uterus

7 · WHY AN ANIMAL BLEEDS · WHAT FOLLOWS

Haemorrhagic diathesis — 4 groups

Group	Examples
Vessel wall	Vitamin C deficiency, endotoxaemia, anthrax , classical swine fever , viral vasculitis, purpura haemorrhagica
Platelets	Thrombocytopenia — ehrlichiosis , bracken fern , immune-mediated, DIC
Clotting factors	Sweet clover (dicoumarol) , warfarin rodenticide, vitamin K deficiency , liver failure, haemophilia A (factor VIII)
Combination	DIC , snake envenomation, aflatoxicosis

Effects depend on **rate**, **volume** and **site**. Rapid loss of **more than one-third of blood volume** is fatal; the same loss slowly is tolerated. A small bleed into the **brain stem or pericardium** kills, a large one into muscle does not.

Fate of shed blood — RBC lysis → haemoglobin → **haemosiderin (iron+, Perls blue)** and **haematoidin (iron-free, golden)**. Colour march of a bruise: red → blue → green → yellow-brown.

PM ARTEFACTS THAT MIMIC HAEMORRHAGE **Hypostatic congestion** — blood settles in the down side, uniform, gravity-dependent, no swelling. **Haemoglobin imbibition** — red staining of intima and endocardium after death. Neither is an antemortem lesion.

8 · OEDEMA — THE 5 MECHANISMS

Mechanism	Classic cause
1 ↑ Hydrostatic pressure	CHF, portal hypertension, venous obstruction
2 ↓ Oncotic pressure (hypoalbuminaemia)	Liver failure, protein-losing nephropathy/enteropathy, parasitism, starvation
3 Lymphatic obstruction	Lymphangitis, tumour, filariasis
4 ↑ Permeability	Inflammation, toxins, anaphylaxis
5 Na⁺/water retention	Renal disease, aldosterone

Anasarca generalised SC **Ascites** peritoneum **Hydrothorax**
Hydropericardium **Bottle jaw** submandibular

INDIA **Bottle jaw** = hypoalbuminaemic oedema of **haemonchosis** & **fasciolosis**.

9 · OEDEMA — WHAT YOU ACTUALLY SEE

- Gross:** tissue swollen and doughy, **pits on pressure**; cut surface **wet, glistening, drips clear straw fluid**; subcutis gelatinous
- Micro:** pale **eosinophilic homogeneous** fluid separating fibres, widened interstitium, dilated lymphatics
- Pulmonary oedema** — heavy wet lungs, **froth in trachea and bronchi**, alveoli filled with pink fluid; the animal drowns in its own plasma
- Cerebral oedema** — flattened gyri, narrowed sulci, **coning of the cerebellum** into the foramen magnum

THE DISTINCTION THAT ANSWERS MOST QUESTIONS

Inflammatory oedema = EXUDATE (high protein, cells, clots, Rivalta +). **Non-inflammatory oedema = TRANSUDATE** (low protein, no clot, Rivalta -).

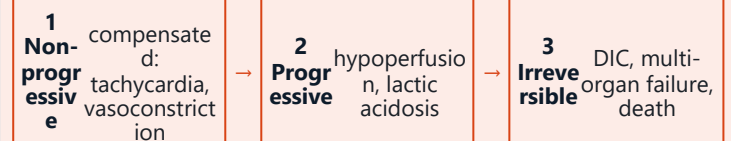
10 · TRANSUDATE VS EXUDATE ★★★

Feature	Transudate	Exudate
Cause	Haemodynamic	Inflammatory
Specific gravity	< 1.012	> 1.018
Protein	< 2.5 g/dl	> 3.0 g/dl
Nucleated cells	< 1000/μl	> 5000/μl
Look	Clear, watery, straw	Turbid, may be purulent
Clotting	Does NOT clot (no fibrinogen)	Clots
Rivalta test	Negative	Positive

TRAP A **modified transudate** lies between (SG 1.012–1.025) — classic in **right-sided heart failure** and hepatic disease.

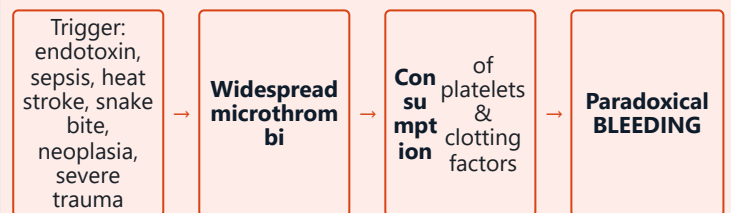
11 · SHOCK — CIRCULATORY COLLAPSE

Type	Mechanism	Example
Hypovolaemic	↓ blood volume	Haemorrhage, burns, severe diarrhoea, dehydration
Cardiogenic	Pump failure	Myocarditis, tamponade, arrhythmia
Distributive	Massive vasodilation	Septic / endotoxic (Gram-negative LPS), anaphylactic , neurogenic



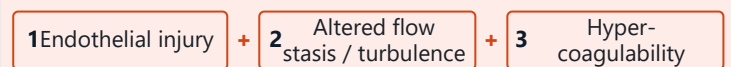
Shock lesions at PM: pulmonary oedema & congestion, "**shock kidney**" (acute tubular necrosis), gut mucosal congestion/necrosis, adrenal cortical depletion, DIC microthrombi.

12 · DIC — "CONSUMPTION COAGULOPATHY"



Lab: ↓platelets · ↑PT & APTT · ↓fibrinogen · ↑FDPs / **D-dimer** · **schistocytes** (fragmented RBCs) on smear. **PM:** petechiae/ecchymoses everywhere + fibrin microthrombi in glomeruli, lung, liver.

13 · THROMBOSIS — VIRCHOW'S TRIAD



Endothelial injury = the single most important factor (heart, arteries); **stasis** dominates in **veins**.

Fate (4): propagation · embolisation · dissolution (plasmin) · **organisation & recanalisation**.

14 · NAMING A THROMBUS

By site	By composition
Mural — on a heart or large-vessel wall	Pale (white) — platelet + fibrin, arterial, fast flow
Valvular — vegetative endocarditis	Red — RBC-rich, venous, slow flow
Occlusive — blocks the lumen	Mixed (laminated) — commonest, shows lines of Zahn
Saddle — straddles a bifurcation	Hyaline — capillary, in DIC
Cardiac / arterial / venous / capillary	

Effects: ischaemia → **infarction** · congestion and oedema behind a venous thrombus · embolism · **septic** spread from an infected thrombus.

15 · THROMBUS VS POST-MORTEM CLOT ★★★

	Thrombus (antemortem)	PM clot
Surface	Dull, dry, granular	Shiny, moist
Feel	Firm, friable	Rubbery, gelatinous
Attachment	Firmly attached	Lifts out freely
Shape	Does not fit vessel	Perfect cast of vessel
Lines of Zahn	PRESENT	Absent
Layers	—	"Chicken fat" over "currant jelly"

DEFINE **Lines of Zahn** = alternating pale platelet-fibrin and dark RBC layers — proof the clot formed in **flowing blood**, i.e. during life.

16 · EMBOLISM

Thromboembolism ~99% · **fat** (long-bone fracture) · **air/gas** (IV injection, bends) · **septic** (valvular endocarditis → embolic nephritis) · **parasitic** (*Strongylus vulgaris* — cranial mesenteric a. of horse) · neoplastic · amniotic fluid · foreign body.

DIRECTION RULE **Venous** emboli lodge in the **LUNG**. **Arterial/left-heart** emboli go to brain, kidney, spleen, limbs. **Paradoxical** embolus crosses a septal defect. **Saddle thrombus** at the aortic trifurcation = classic **cat** with HCM.

17 · INFARCT — RED VS WHITE

	Red (haemorrhagic)	White (pale/anaemic)
Tissue	Loose, spongy	Solid, compact
Supply	Dual circulation	End-arterial
Occlusion	Often venous	Arterial
Organs	Lung, intestine, testis	Kidney, heart, spleen

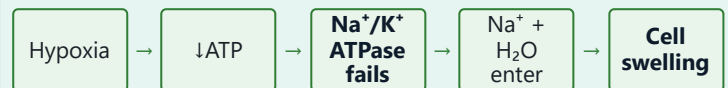
Shape = **wedge/cone**, apex toward the occluded vessel, base at capsule. Necrosis = **coagulative**, except the **BRAIN** which **liquefies**. Old infarct = pale depressed **scar**.

18 · WHAT DECIDES WHETHER AN INFARCT FORMS

- Type of blood supply** — a **dual** supply (lung, liver) protects; an **end-artery** (kidney, spleen, retina) does not
- Rate of occlusion** — slow occlusion lets **collaterals** open, so no infarct forms
- Vulnerability to hypoxia** — **neuron 3–5 min** < myocardium 20–30 min < fibroblast hours
- Oxygen content of the blood** — anaemia and heart failure make infarction more likely

TIMELINE OF A WHITE INFARCT 0–24 h pale, swollen, with a **red hyperaemic rim** → 2–4 days yellow-white, sharply demarcated, leucocytes at the margin → 1–2 weeks granulation tissue moves in → weeks to months a **depressed pale fibrous scar**.

19 · REVERSIBLE CELL INJURY

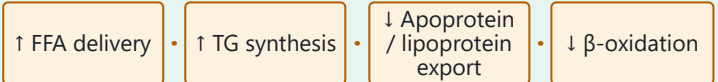


- Cloudy swelling** — **earliest & commonest**; organ pale, swollen, heavy, dull "parboiled", cut surface bulges
- Hydropic/vacuolar** — larger clear vacuoles
- Hyaline** — glassy pink homogeneous (Russell bodies, hyaline casts)
- Zenker's (waxy)** — of **skeletal muscle**: white muscle disease, blackleg, capture myopathy
- Mucoid/mucinous** degeneration

POINT OF NO RETURN **Membrane damage** + **mitochondrial permeability transition** + massive **Ca²⁺ influx**. Clinically = leaked enzymes (ALT, AST, CK, LDH).

20 · FATTY CHANGE & GLYCOGEN ★

FATTY CHANGE (steatosis) — abnormal **triglyceride** accumulation, chiefly in the **LIVER** (also heart, kidney).

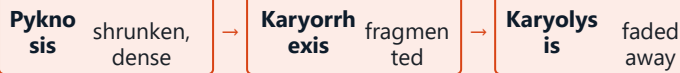


- Causes:** hypoxia · **ketosis of crossbred cows** · **pregnancy toxemia** of ewes/does · **feline hepatic lipidosis** · **aflatoxin** · CCl₄, phosphorus · protein-energy malnutrition · diabetes mellitus · copper & choline deficiency
- Gross:** liver **enlarged, yellow, greasy, soft, rounded borders**, greasy knife; cut surface bulges; **FLOATS in water/formalin**
- Micro (H&E):** sharply punched-out **clear round vacuoles** displacing the nucleus — the fat is **dissolved out** by paraffin processing
- To prove it is fat:** **FROZEN section** + **Sudan III / IV** (orange-red), **Oil Red O** (red), **Osmium tetroxide** (black)

FATTY CHANGE VS FATTY INFILTRATION **Fatty change** = fat inside **parenchymal cells** (pathological). **Fatty infiltration** = mature **adipocytes between** cells (obesity, largely physiological).

GLYCOGEN Excess in **diabetes mellitus**, **hyperadrenocorticism/steroids** ("steroid hepatopathy" of the dog), glycogen storage disease. Stain **PAS** → **magenta**; confirm by **diastase digestion** — glycogen **disappears**, so PAS+ / diastase-labile. Fix in **alcohol or Carnoy's**, because aqueous fixatives dissolve glycogen.

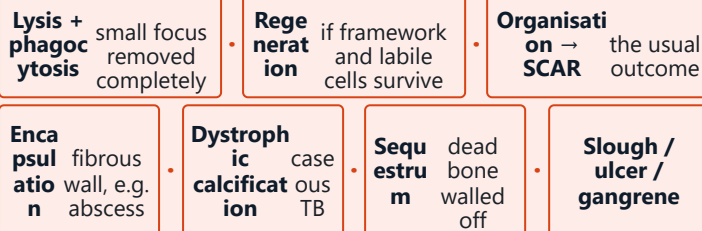
21 · NECROSIS



Earliest cytoplasmic sign = **increased eosinophilia** (loss of basophilic RNA).

Type	Key point
Coagulative	Commonest ; ischaemia; protein denaturation; architecture preserved ; kidney, heart, spleen
Liquefactive	BRAIN + pyogenic bacteria → abscess
Caseous	Cheesy, dry, crumbly ; TB, caseous lymphadenitis; architecture lost
Fat	Enzymatic (pancreatic lipase) → chalky-white saponification; also traumatic; abdominal fat necrosis of cattle
Fibrinoid	Bright pink in vessel walls ; immune-complex vasculitis

22 · FATE OF NECROTIC TISSUE



Markers of necrosis in blood: ALT and SDH liver · AST and CK muscle · ALP and GGT biliary · amylase and lipase pancreas · troponin I myocardium.

23 · APOPTOSIS VS NECROSIS ★★★

	Apoptosis	Necrosis
Nature	Programmed , physiological	Pathological
Extent	Single scattered cells	Groups of cells
Cell	Shrinks	Swells
Membrane	Intact (blebs)	Ruptures
Energy	ATP required	ATP-independent
Inflammation	ABSENT	PRESENT
Enzymes	Caspases	Lysosomal hydrolases
DNA gel	LADDER (180–200 bp)	SMEAR
Marker	Phosphatidylserine flip → Annexin V , TUNEL	Enzyme leakage

NAMED BODY Councilman body = apoptotic hepatocyte (infectious canine hepatitis, viral hepatitis).

24 · NECROSIS VS PM AUTOLYSIS ★★★

	Necrosis	Autolysis
When	During life	After death
Distribution	Focal , sharply limited	Diffuse, uniform , whole body
Demarcation zone	Present	Absent
Inflammation	Present (vital reaction)	Absent

FIELD RULE No vital reaction = the animal was already dead. Autolysis is fastest in India's heat, and fastest in **gut mucosa, pancreas, adrenal medulla, gall bladder**.

25 · GANGRENE — 3 TYPES

	Dry	Moist (wet)	Gas
Vessels	Artery blocked, vein open	Both blocked	Any + toxin
Tissue	Dry, shrunken, mummified , black	Swollen, soft, foul	Crepitant , emphysematous
Demarcation	Clear	Absent	Absent
Site	Ear tip, tail, teat, foot	Lung, gut, udder, uterus	Muscle
Cause	Ergot , fescue, frostbite, tight bandage	Aspiration pneumonia, gangrenous mastitis, torsion	Cl. chauvoei (blackleg), <i>Cl. septicum</i> , <i>Cl. perfringens</i>

26 · PIGMENTS — THE STAIN IS THE ANSWER

EXOGENOUS — **anthracosis** (carbon, lung + bronchial nodes) · silicosis · tattoo · **lead line** on gums · **carotenoid** yellow fat of **Jersey/Guernsey** (normal) · **tetracycline** yellow teeth, **UV-fluorescent**.

Endogenous	Nature / Stain
Haemosiderin	Iron+, golden-brown; Perls' Prussian blue → BLUE
Haematoidin	Iron-FREE , golden crystals in centre of old haemorrhage; Perls' NEGATIVE
Bilirubin	Icterus. van den Bergh direct = obstructive/hepatic; indirect = haemolytic
Formalin pigment (acid haematin)	ARTEFACT of unbuffered formalin; brown-black, birefringent . Prevented by 10% neutral buffered formalin
Parasitic haematin	Black liver in fasciolosis
Melanin	Masson-Fontana +, bleached by H ₂ O ₂ , Perls' negative . Melanosis = harmless congenital black spots in lung/meninges of calves & pigs
Lipofuscin	" Wear-and-tear " ageing pigment → brown atrophy of heart/liver; PAS+, acid-fast
Porphyrin	Congenital erythropoietic porphyria — " pink tooth ", pink bone, red UV fluorescence ; Holstein

27 • ICTERUS (JAUNDICE) — 3 TYPES ★

	Pre-hepatic (haemolytic)	Hepatic (toxic)	Post-hepatic (obstructive)
Bilirubin type	Indirect (unconjugated)	Both	Direct (conjugated)
van den Bergh	Indirect +	Biphasic	Direct +
Urine bilirubin	Absent (water-insoluble)	Present	Present
Faeces	Dark	Normal/pale	Pale, clay-coloured
Example	Babesiosis, anaplasmosis, copper toxicity, neonatal isoerythrolysis	Aflatoxin, leptospirosis, CCl ₄	Fasciolosis, gall stone, ascarid in bile duct, tumour

INDIA Jaundice in a crossbred cow in the tick season = think **babesiosis / theileriosis** before liver disease.

28 • CALCIFICATION ★★★

	Dystrophic	Metastatic
Tissue	Dead / dying	Normal, living
Serum Ca	NORMAL	RAISED
Spread	Local	Widespread
Cause	Caseous TB, fat necrosis, old infarct, dead parasites, atheroma	Vitamin D toxicity, enzootic calcinosis (<i>Solanum malacoxylon</i> , <i>Trisetum</i>), hyperparathyroidism, CRF, bone lysis

METASTATIC SITES Lung • Kidney • Gastric mucosa • Systemic arteries • Pleura — the tissues that **lose acid** and are internally alkaline.

STAINS von Kossa → **BLACK** (detects phosphate). **Alizarin Red S** → **orange-red** (detects calcium itself).

Calcinosis circumscripta — nodules on pressure points, large dogs. **Calcinosis cutis** — hyperadrenocorticism. **Gout** = urate, NOT calcification.

29 • AMYLOIDOSIS

- **β-pleated sheet**, non-branching fibrils **7.5–10 nm + P component**
- **AA (secondary/reactive)** from **SAA** — **commonest in animals**; chronic suppurative, TB, hyperimmunised horses. **AL (primary)** = Ig light chain
- Order: **KIDNEY (glomerulus) first** → liver → spleen → adrenal. Renal amyloid → **protein-losing nephropathy**
- **Sago spleen** = nodular/follicular • **Lardaceous (bacon) spleen** = diffuse red pulp

STAINS ★★★ Congo red → salmon-red in ordinary light, **APPLE-GREEN BIREFRINGENCE** in **polarised** light. **Methyl/crystal violet** → metachromatic pink-red. **Thioflavin T** → fluorescent. Gross: **Lugol's iodine + dilute H₂SO₄** → mahogany brown → blue-violet.

30 • PHOTSENSITIZATION — 4 TYPES

Type	Mechanism & agent
I Primary	Preformed agent eaten: hypericin (St John's wort), fagopyrin (buckwheat), phenothiazine
II	Aberrant pigment metabolism — congenital erythropoietic porphyria, bovine protoporphyria
III Hepatogenous	COMMONEST . Liver damage → phyloerythrin (chlorophyll product) not excreted in bile. Lantana camara (lantadene A) , <i>Tribulus</i> , <i>Panicum</i> , sporidesmin (facial eczema, <i>Pithomyces chartarum</i>)
IV	Idiopathic

LESION RULE Only **WHITE (unpigmented)**, **hairless**, **sun-exposed** skin — muzzle, eyelid, ear, teat, perineum, coronet. Sunburn hits pigmented skin too; photosensitization does not.

31 • DISTURBANCES OF GROWTH

Agenesis	primordium never formed
Aplasia	primordium present, no development
Hypoplasia	incomplete development
Atrophy	shrinkage of a once-normal organ
Hypertrophy	↑ cell SIZE
Hyperplasia	↑ cell NUMBER
Metaplasia	reversible swap of one mature cell type for another
Dysplasia	disordered growth — pre-neoplastic
Anaplasia	loss of differentiation — malignancy

TRAPS ① **Permanent** cells (cardiac, skeletal muscle, neuron) can only **hypertrophy**, never hyperplase. ② **Squamous metaplasia** = classic **vitamin A deficiency**. ③ Atrophy causes: **disuse, denervation, ischaemia, nutritional, pressure, loss of endocrine stimulus, ageing**. ④ **Serous atrophy of fat** — gelatinous coronary-groove & marrow fat = cachexia.

32 • ADAPTATIONS — NAMED EXAMPLES

Adaptation	Physiological	Pathological
Hypertrophy	Pregnant uterus, working muscle	Cardiac hypertrophy in valvular disease • bladder wall in urethral obstruction
Hyperplasia	Mammary gland in pregnancy, liver after partial resection	Goitre (iodine deficiency) • endometrial and prostatic hyperplasia • skin callus
Atrophy	Thymus, involuting uterus	Disuse after fracture • denervation muscle atrophy • hydronephrosis (pressure) • starvation
Metaplasia	—	Squamous metaplasia in vitamin A deficiency and chronic irritation • osseous metaplasia in old scars and lung • myeloid metaplasia of the spleen

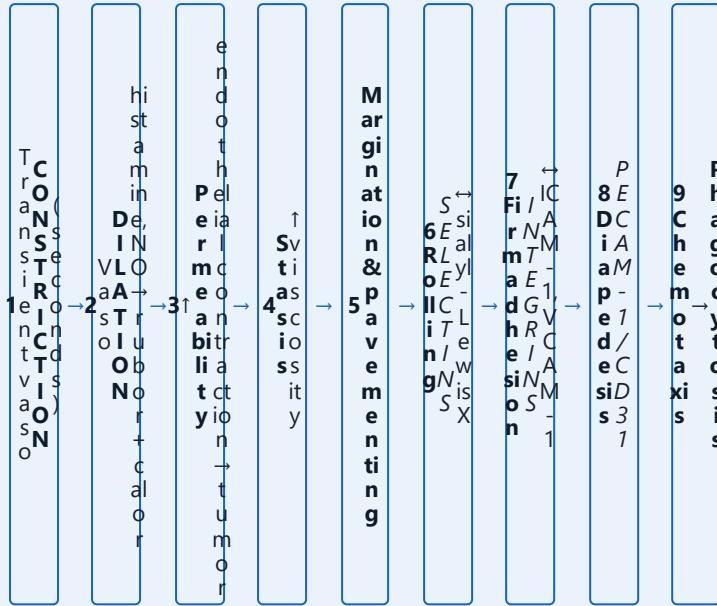
DIRECTION OF THINKING Adaptation is **reversible and purposeful** — remove the stimulus and the tissue returns. **Dysplasia** is where control begins to fail; **neoplasia** is where it is lost permanently.

33 · INFLAMMATION — SIGNS & VASCULAR SEQUENCE

Rubor red **Calor** heat **Tumor** swelling **Dolor** pain

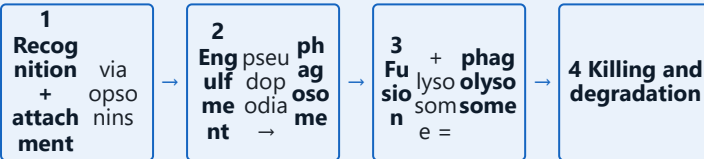
Functio laesa loss of function

Celsus gave the first four; **Virchow** added *functio laesa*.



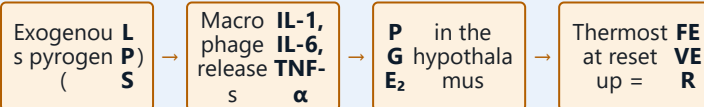
LEARN AS SETS Chemotaxins: C5a · LTB₄ · IL-8 · bacterial *N*-formyl-methionyl peptides. Opsonins: IgG Fc · C3b · collectins. **Killing:** the MPO–H₂O₂–halide system is the most efficient.

34 · PHAGOCYTOSIS AND FEVER



Oxygen-dependent killing: NADPH oxidase **respiratory burst** → O₂⁻ → H₂O₂ → MPO–H₂O₂–halide → HOCl (most potent).

Oxygen-independent: lysozyme, lactoferrin, defensins.



DEFECTIVE PHAGOCYTOSIS Chédiak–Higashi — faulty lysosomal fusion, **giant granules**. BLAD — CD18 integrin defect, so leucocytes cannot adhere: **marked neutrophilia but pus does not form**.

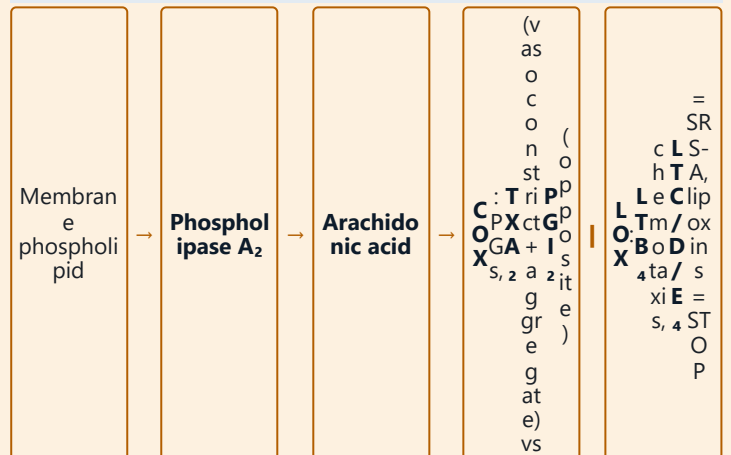
35 · INFLAMMATORY CELLS

Cell	Arrives	Signature
Neutrophil	FIRST 6–24 h	Acute, pyogenic; forms pus . Heterophil in birds, reptiles, rabbits
Macrophage	24–48 h	Chronic ; from monocyte; antigen presentation; → epithelioid cell
Eosinophil	—	Parasites & allergy ; major basic protein; IL-5
Mast/basophil	—	Histamine, heparin; Type I ; metachromatic with toluidine blue
Plasma cell	Late	Cartwheel nucleus; Russell bodies = Ig globules; stuffed = Mott cell

GIANT CELLS & GRANULOMA Langhans — nuclei in a **horseshoe at the periphery** (TB). **Foreign-body** — nuclei **random**. **Touton** — ring of nuclei, foamy (fat). **Granuloma** = focal collection of **epithelioid macrophages** ± giant cells ± lymphocyte cuff ± caseation ± fibrous capsule.

36 · CHEMICAL MEDIATORS — BY JOB

Effect	Mediators
Vasodilation	Histamine, PGI ₂ , PGD ₂ , PGE ₂ , NO
↑ Permeability	Histamine (first, immediate), serotonin, bradykinin , C3a & C5a (anaphylatoxins), LTC₄/D₄/E₄ , PAF, substance P
Chemotaxis	C5a , LTB ₄ , IL-8 , bacterial peptides
Fever	IL-1 , IL-6 , TNF-α (endogenous pyrogens) → PGE₂ at hypothalamus
Pain	Bradykinin , PGE ₂ , substance P
Tissue damage	Lysosomal enzymes, ROS, NO



DRUG TRAP NSAIDs block COX only. Corticosteroids block phospholipase A₂ — therefore **BOTH** pathways.

Complement: classical = antigen–antibody (C1) · alternative = endotoxin (C3) · **lectin** = MBL. **C3a/C5a** anaphylatoxins · **C5a** chemotaxis · **C3b** opsonin · **C5b-9** = MAC.

37 · OUTCOMES OF ACUTE INFLAMMATION

Complete resolution — little damage, labile tissue	Healing by fibrosis — much damage, or permanent cells	Abscesses — pyogenic organisms	Chronic inflammation — the agent persists
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Beneficial — dilution of toxin, delivery of antibody and drugs, fibrin traps organisms, phagocytosis, immunity, framework for repair.

Harmful — pain and loss of function, digestion of normal tissue, swelling inside a closed space (**brain, larynx, pericardium**), fibrosis and **adhesions**, hypersensitivity, chronic **amyloidosis**.

38 · CLASSIFICATION OF INFLAMMATION

By duration	By exudate	By distribution
Peracute · Acute (neutrophil, vascular, days)	Serous · Catarrhal · Fibrinous · Suppurative	Focal · Multifocal ·
Subacute · Chronic (macrophage, lymphocyte, plasma cell, fibrosis , weeks+)	Haemorrhagic · Necrotising · Granulomatous	Diffuse · Locally extensive

Chronic inflammation arises by: persistence of the agent (TB, Johne's, foreign body) · repeated acute attacks · autoimmunity · low-grade toxicity.

GRANULOMA VS GRANULATION TISSUE ★ **Granuloma** = a **chronic inflammatory nodule** of epithelioid macrophages. **Granulation tissue** = **new capillaries + fibroblasts** of **healing**. Examiners swap these deliberately.

Named chronic lesions: **tubercle** (TB) · **gumma** · **abscess** (chronic, walled by pyogenic membrane) · **fistula** (two open ends) vs **sinus** (one blind end) · **ulcer** (loss of surface epithelium) vs **erosion** (partial thickness).

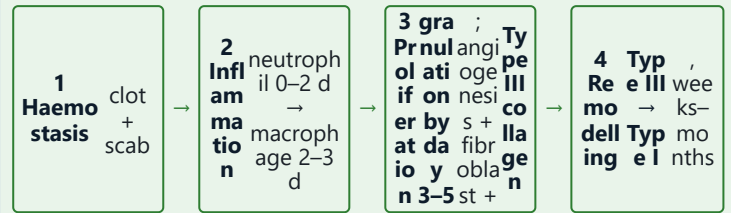
39 · TYPES OF EXUDATE + SYSTEMIC EFFECTS

Exudate	Classic example
Serous	Watery, low protein — blisters, early burns
Fibrinous	" Bread-and-butter " pericarditis. Croupous peels off easily; diphtheritic is adherent and leaves an ulcer
Suppurative	Pus = dead neutrophils. Abscess localised · phlegmon/cellulitis diffuse · empyema in a cavity
Catarrhal	Mucus-rich, mucous membranes
Haemorrhagic	Anthrax, HS, swine fever

SYSTEMIC EFFECTS Fever · **leucocytosis** with **LEFT SHIFT** — **regenerative** (mature > immature) vs **DEGENERATIVE** (immature > mature = **bad prognosis**) · leucopenia in viral disease & overwhelming sepsis · **acute-phase proteins:** **CRP** → **dog, SAA** → **cat, haptoglobin** → **cattle, fibrinogen** → **horse** · ↑ESR · weight loss · amyloidosis if chronic.

40 · WOUND HEALING

	Primary (first) intention	Secondary intention
Wound	Clean, sutured, edges apposed	Large, open, tissue lost
Granulation tissue	Scanty	Abundant
Contraction	Negligible	Marked — myofibroblasts
Scar	Thin, neat	Large, disfiguring



• **Granulation tissue** = new capillaries + fibroblasts + loose ECM + inflammatory cells. Red, granular, bleeds easily. **NOT the same as granuloma.**

• **Tensile strength** ~10% at 1 week → **maximum only 70–80% by 3 months. Never 100%.**

• **Vitamin C** — hydroxylation of proline & lysine · **Zinc** — epithelialisation · **Copper** — lysyl oxidase

GROWTH FACTORS **VEGF** = most potent **angiogenic**. **TGF-β** = most potent **fibrogenic** (↑collagen, ↓collagenase). **PDGF** = fibroblast & smooth-muscle migration. **EGF/TGF-α** = epithelialisation.

COMPLICATIONS & CELL TYPES Dehiscence · keloid · **exuberant granulation tissue** = "**PROUD FLESH**", classic on the **distal limb of the HORSE** · contracture · adhesions. Delayed by **infection (most important)**, foreign body, movement, poor blood supply, **corticosteroids**, protein/vit-C/zinc deficiency.

Labile (skin, gut, marrow) always divide · **Stable** (liver, kidney, pancreas) divide on demand · **Permanent** (neuron, cardiac & skeletal muscle) **heal by SCAR only.**

41 · HEALING OF SPECIAL TISSUES

Tissue	How it heals
Bone	Haematoma → procallus → soft (fibrocartilaginous) callus → bony callus → remodelling. The only tissue that heals by true regeneration with original tissue
Peripheral nerve	Wallerian degeneration distal to the cut; axon regrows ~1 mm/day along the Schwann-cell tube; a blocked stump forms a traumatic (amputation) neuroma
CNS	No regeneration. Heals by gliosis (astrocytes) — the CNS "scar"
Liver	Excellent regeneration if the reticulin framework survives; if not → cirrhosis (regenerative nodules + fibrosis)
Cardiac muscle	Fibrous scar only — no new myocytes

42 · IMMUNOPATHOLOGY — COOMBS & GELL

Type	Mediator	Examples
I Immediate / anaphylactic	IgE on mast cells → histamine, minutes	Anaphylaxis, atopy, flea-bite dermatitis, urticaria, milk allergy in Jersey cows
II Cytotoxic	IgG/IgM vs cell-surface antigen + complement / ADCC	Neonatal isoerythrolysis (foal, piglet), AIHA, myasthenia gravis, pemphigus
III Immune complex	Ag–Ab complexes → complement → neutrophils	Arthus (local), serum sickness (systemic), glomerulonephritis, "blue eye" of ICH , purpura haemorrhagica, equine recurrent uveitis
IV Delayed / cell-mediated	T lymphocytes + macrophages, 24–72 h, no antibody	Tuberculin & Johnin test , contact dermatitis, granuloma, graft rejection

SHOCK ORGAN ★ Dog → LIVER (hepatic veins) · horse, pig, ruminant, cat → LUNG.

Immunodeficiency — primary: SCID of Arabian foals · BLAD (Holstein, CD18) · Chédiak–Higashi (Hereford cattle, Persian cat — giant granules) · selective IgA deficiency (German Shepherd).

Secondary (commoner): FAILURE OF PASSIVE TRANSFER — the commonest of all; serum IgG < **400 mg/dl** in calf/foal · BVD, CDV, FeLV/FIV · malnutrition · corticosteroids and stress.

Autoimmunity: SLE (**LE cell**, **ANA+**) · **pemphigus foliaceus** = commonest autoimmune skin disease of the dog, **acantholysis** → **acantholytic cells** · autoimmune thyroiditis · myasthenia gravis (anti-AChR).

43 · WHY AUTOIMMUNITY HAPPENS

- **Release of sequestered antigen** — lens, sperm, thyroglobulin, CNS myelin never met the immune system in fetal life
- **Molecular mimicry** — a microbial antigen resembles a host antigen
- **Alteration of self antigen** by drug, virus or hapten
- **Loss of regulatory T-cell control** or polyclonal lymphocyte activation
- **Genetic predisposition** — MHC haplotype and breed

TESTS TO REMEMBER Coombs test — antibody on red cells, for **AIHA** and **neonatal isoerythrolysis**. **ANA** — for **SLE**. **LE cell** — a neutrophil that has engulfed a nucleus. **Direct immunofluorescence** of skin biopsy — for **pemphigus**. **Tuberculin / Johnin** — Type IV delayed reaction.

44 · 60-SECOND BLANK-FILL DRILL

- Lines of Zahn prove the clot formed **during life**
- Transudate SG is below **1.012**; Rivalta is positive for **exudate**
- Heart-failure cells = **haemosiderin-laden alveolar macrophages**
- Nutmeg appearance = **liver** in chronic passive congestion
- Commonest necrosis **coagulative**; brain shows **liquefactive**
- Nuclear sequence **pyknosis** → **karyorrhexis** → **karyolysis**
- Apoptosis DNA = **ladder**; necrosis DNA = **smear**; enzymes = **caspases**
- Blackleg gangrene organism = **Clostridium chauvoei**
- Perls' stains **haemosiderin** blue; haematoidin is **iron-free**
- Formalin pigment is avoided by **10% neutral buffered formalin**
- von Kossa = **black**; alizarin red S = **orange-red**
- Serum Ca is **normal** in dystrophic, **raised** in metastatic calcification
- Hepatogenous photosensitization agent = **phylloerythrin**
- Lantana toxin **lantadene A**; St John's wort **hypericin**
- Squamous metaplasia = **vitamin A** deficiency
- Functio laesa added by **Virchow**; first four by **Celsus**
- First cell in acute inflammation = **neutrophil** (**heterophil** in birds)
- Rolling uses **selectins**; firm adhesion uses **integrins**
- Most potent angiogenic factor **VEGF**; most fibrogenic **TGF-β**
- Corticosteroids inhibit **phospholipase A₂**; NSAIDs inhibit **COX**
- Wound tensile strength peaks at **70–80%** by **3 months**
- Proud flesh = **exuberant granulation tissue** of the **horse**
- Amyloid = **apple-green birefringence**, Congo red, polarised light
- Commonest animal amyloid **AA**, from **serum amyloid A**
- Tuberculin test = type **IV** hypersensitivity
- Shock organ: dog **liver**, horse **lung**
- Commonest immunodeficiency in farm animals = **failure of passive transfer**
- Fatty liver **floats** in water; fat is shown by **Sudan III / Oil red O** on a **frozen** section
- Glycogen is **PAS positive** and removed by **diastase**; fix in **alcohol/Carnoy's**
- Schistocytes and ↑ D-dimer indicate **DIC**
- Obstructive jaundice gives **direct** (conjugated) bilirubin and **pale** faeces
- Nerve axons regenerate at about **1 mm per day** after **Wallerian** degeneration
- The CNS heals by **gliosis**, not by fibrosis
- Bone heals through **procallus** → **soft callus** → **bony callus**
- A tract with two open ends is a **fistula**; with one blind end a **sinus**
- Langhans giant cell nuclei lie in a **horseshoe at the periphery**
- Endotoxic shock is a **distributive** shock caused by Gram- **negative** LPS
- A wedge-shaped infarct has its apex toward the **occluded vessel**
- Neurons die of hypoxia in about **3–5 minutes**
- Sweet clover poisoning bleeds through **dicoumarol**, a vitamin **K** antagonist
- The respiratory burst enzyme is **NADPH oxidase**
- BLAD cattle show neutrophilia but **cannot form pus** (CD18 defect)
- Chédiak–Higashi leucocytes contain **giant granules**
- Coombs test detects **antibody bound to red cells**; ANA is positive in **SLE**
- Iodine-deficiency goitre is an example of **hyperplasia**
- Hydronephrosis causes **pressure** atrophy of the kidney
- Granuloma is a lesion of **chronic inflammation**; granulation tissue is a lesion of **healing**